

Sales Data Analysis

Model Evaluation & Findings Report

Program	CloudExify Data Science Internship — Summer 2026
Module	Month 1 · Project 1
Domain	Exploratory Data Analysis (EDA)
Toolset	Python · pandas · NumPy · matplotlib
Report Date	July 2026

1. Executive Summary

This report documents the exploratory data analysis performed on a full-year retail sales dataset as part of Project 1 of the CloudExify Data Science internship. The workflow covered data ingestion, cleaning, descriptive statistics, and visual analysis across products, categories, and regions, using a standard pandas / NumPy / matplotlib pipeline in a Jupyter notebook environment.

The cleaned dataset comprised 485 valid sales records spanning the full 2025 calendar year, generating a total tracked revenue of Rs 12,423,188. The analysis identifies the Laptop as the top revenue-generating product and August 2025 as the strongest sales month, providing a clear basis for further business and modeling decisions.

2. Objective & Methodology

The objective of this project was to build an end-to-end sales analysis pipeline capable of cleaning raw transactional data and surfacing actionable revenue insights through statistical summaries and visualizations. The methodology followed five stages:

Stage	Description
Data Ingestion	Loaded raw CSV sales records into a pandas DataFrame; inspected shape, dtypes, and null distribution.
Data Cleaning	Removed rows with missing values, dropped duplicate records, and converted the Date column to a proper datetime type.
Descriptive Statistics	Computed total, mean, median, and standard deviation of sale amounts; tabulated product frequency counts.
Aggregation	Grouped revenue by Product, Category, and Region to surface top performers and distribution patterns.
Visualization	Produced a top-10 product bar chart, a monthly revenue trend line plot, a regional pie chart, and a category comparison chart.

3. Dataset Overview

Metric	Value
Total valid records	485
Date range	2025-01-01 to 2025-12-31 (full calendar year)
Total revenue	Rs 12,423,188
Average sale value	Rs 25,615
Median sale value	Rs 25,970

The near-equal average (Rs 25,615) and median (Rs 25,970) sale values indicate a roughly symmetric revenue distribution across transactions, with no extreme skew from outlier sales.

4. Product Performance

Grouping total revenue by product identifies the five highest-earning items in the catalog. The Laptop category leads by a clear margin, followed by Webcam, Headphones, Charger, and Keyboard.

Rank	Product	Revenue (Rs)
1	Laptop	1,627,608
2	Webcam	1,376,670
3	Headphones	1,355,870
4	Charger	1,287,786
5	Keyboard	1,238,179

Together, the top five products account for Rs 6,886,113 — roughly 55% of total tracked revenue — indicating that revenue is concentrated in a relatively small set of high-performing items.

5. Temporal Trend

Aggregating revenue by calendar month surfaces seasonal purchasing patterns across the year. **August 2025** emerged as the single best-performing month by total revenue, suggesting a seasonal demand peak that could inform inventory planning and promotional timing for future cycles.

6. Key Findings

- Revenue distribution is approximately symmetric, with mean and median sale values within 1.4% of each other.
- Laptop is the single highest revenue-generating product, ahead of Webcam and Headphones.
- The top 5 products by revenue account for roughly half of total tracked sales, indicating moderate product concentration.
- August 2025 was identified as the peak sales month across the full 12-month window.
- The dataset required standard cleaning (missing values and duplicate rows) before analysis, consistent with typical raw transactional exports.

7. Recommendations

- Prioritize inventory and marketing spend around top-performing products (Laptop, Webcam, Headphones) heading into peak months.
- Investigate the drivers behind the August revenue peak (seasonality, promotions, restocking cycles) to replicate it in other months.
- Extend the analysis with year-over-year data once available, to distinguish seasonal effects from one-off spikes.
- Consider a follow-up cohort or regional deep-dive to identify whether revenue concentration in top products holds consistently across all regions.

8. Conclusion

The Project 1 pipeline successfully transformed a raw, imperfect sales dataset into a clean, analyzable form and produced clear, actionable insights on product performance, revenue distribution, and seasonal trends. The findings establish a solid foundation for the predictive modeling and deeper statistical work expected in subsequent project phases of the internship.